Donghee (Joy) Chae

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Research Interests

To engineer immune-instructive materials for cancer and inflammatory disease, and to advance science communication that keeps such therapies intelligible to the public they are built for.

- **Stem cell-based immunomodulation.** Immunosuppressive capacity of mesenchymal stromal cells; efferocytosis-driven immunosuppression in monocytes and their macrophage progeny; iPSC-derived model systems.
- **Nanomaterials.** Radiation-responsive scintillating nanoparticles for X-ray-triggered uncaging; non-viral polymer nanocomplexes for CAR-TAM engineering.
- **Bioinformatics and *in silico* modeling.** HER2-ADC response prediction from endocytic trafficking biology; RNA-seq analysis of therapy-induced tumor immune remodeling; structure-based drug repurposing for radiation-activated payloads.
- **Environmental exposure and immune function.** Pb, PCBs, and PFAS as modifiers of antigen presentation and cellular immune response.

Education

Korea University | Seoul, Republic of Korea — **Mar 2022 – Feb 2027** *(expected)*
B.S. in Health and Environmental Science
B.E. in Biomedical Engineering (dual degree, admitted Nov 2023)

University of Iowa | Iowa City, IA, USA — **Aug 2025 – May 2026**
Exchange Student, Roy J. Carver Department of Biomedical Engineering

Research Experience

Nano-Biomedicine Lab, Korea University (PI: Mikyung Kang) — *Seoul, Korea*
Undergraduate Researcher — **Jun 2026 – Present**

- **Radiation-activated immunotherapy** (KIRAMS and Harvard Medical School collaboration)
  - Synthesized **inorganic nanoparticles** that convert X-ray energy into UV emission to cleave a UV-labile linker.
  - Designed the **in silico drug repurposing** workflow used to identify an IL-2-mimetic docking system.
- **Propagation of MSC-induced immunosuppression to macrophages**
  - Tested whether the immunosuppressive phenotype of **efferocytic macrophages** propagates to bystander macrophages through exosome-mediated transfer
- **HER2 ADC response prediction** (KIST molecular imaging collaboration)
  - Analyzed GEO datasets to prioritize **biomarkers** of **T-DXd** and **T-DM1** response.
  - Tracked antibody internalization and **lysosomal degradation** by immunocytochemistry and confocal imaging
  - Built a **predictive mathematical model** linking trafficking-gene expression to ADC response
- **CAR-TAM drug factory** (Korea University, Kyung Hee University, and Yale Cancer Center collaboration)
  - Developed the **non-viral delivery system** replacing lentiviral vectors, using PEI-plasmid nanocomplexes that deliver CRISPR-Cas9 and an HITI donor for single-step **SIGLEC10 knockout and CAR knock-in**
- **Environmental immunotoxicology**
  - Designed a study asking whether non-cytotoxic **Pb** exposure impairs **phagolysosomal** antigen processing and antigen-specific **$CD8^+$ T-cell activation** in DC2.4 dendritic cells

Ankrum Lab, University of Iowa (PI: James Ankrum) — *Iowa City, IA*
Undergraduate Research Intern — **Aug 2025 – May 2026**

- **Efferocytosis of cytokine pre-licensing of MSCs**
  - Showed that **cytokine pre-licensing of MSCs** elicits the efferocytic response of primary human monocytes, with downstream **suppression of T-cell proliferation**; led manuscript preparation.
  - Validated the five-color monocyte/macrophage phenotyping panel (CD14, CD16, CD86, CD163, CD206) on flow cytometry and performed all FlowJo analysis.
  - Quantified secreted TNF-α, IFN-γ, IL-10, and IL-4 using ELISA
- **Subacute PCB inhalation toxicology**(with Prof. Andrea Adamcakova-Dodd)
  - Ran the murine inhalation exposure study on PCB1254 combined with a Western diet**.**
  - Performed mice necropsy and multi-organ harvest, followed by nucleic acid extraction and PCR
- **Metabolic profiling of mouse islet alpha and beta cells** (with Prof. Yumi Imai)
  - Resolved alpha and beta cell subpopulations by puromycin signal across four mouse islet flow cytometry datasets

ChEEHR Lab, Korea University (PI: Yoon-Hyeong Choi) — *Seoul, Korea*
Undergraduate Researcher — **Mar 2024 – Sep 2024**

- Analyzed post-disaster exposure data from the **Humidifier Disinfectant Incident** for national health risk assessment.
- Curated heavy metal exposure data and performed the epidemiological analysis.

Publications

Under review

[1] **D. Chae**, D. del Bosque Siller, J. N. Liszewski, R. Byrd, M. V. Schrodt, R. M. Behan-Bush, and J. A. Ankrum*, *"Cytokine pre-licensing of mesenchymal stromal cells amplifies the efferocytic immunosuppressive response of monocytes and their macrophage progeny,"* under review (2026).

[2] **D. Chae**, M. Kim, Y. Y. Hwang, C. You, W. Ju, I. Noh*, and M. Kang*, *"Beyond ex vivo CAR-T therapies through in vivo programming and targeted delivery,"* under review (2026).

[3] J. Jeong†, **D. Chae**†, J. Lee, S. Park, M. A. Miller, and M. Kang*, *"Radiation-responsive nanomaterials: mechanistic foundations and emerging frontiers in cancer theranostics,"* under review (2026).

In preparation

[4] A. Adamcakova-Dodd, H.-J. Lehmler, X. Jing, H. Wang, A. Klingelhutz, J. A. Ankrum, **D. Chae**, D. May, and P. S. Thorne*, *"Subacute inhalation exposure to PCB1254 with a Western diet alters metabolic and pulmonary inflammatory responses in mice,"* in preparation.

[5] S. Liu, F. Haider, **D. Chae**, L. Kim, R. Byrd, J. A. Ankrum, Yumi Imai*, *"A role of alpha cell fatty acid metabolism in glucagon secretion,"* in preparation.

[6] J. Gong, H. Kang, J. Kang, J. Kim, Y. Jin, **D. Chae**, J. Jung, and Y. Choi*, *"Health effects of heavy metal exposure in Korean adults: a review of national biomonitoring data,"* in preparation.

†Equal contribution, *Corresponding author

Conference Presentations

[1] **D. Chae** & M. Kang et al., *"Efferocytosis of mesenchymal stromal cells potentiates an immunosuppressive monocyte phenotype that endures after macrophage differentiation,"* **KSBM 2026** (Korean Society for Biomaterials Annual Meeting), Busan, Korea. (2026, September).

[2] S. Liu, F. Haider, **D. Chae**, L. Kim, R. Byrd, J. A. Ankrum, Yumi Imai*, *"A role of alpha cell fatty acid metabolism in glucagon secretion,"* **Midwest Islet Club 2026**, Madison, WI. (2026, May).

[3] **D. Chae**, *"Cytokine pre-licensing of mesenchymal stromal cells and monocyte efferocytosis,"* **Research Open House 2026**, College of Engineering, University of Iowa, Iowa City, IA. (2026, April).

[4] **D. Chae** & J. A. Ankrum et al., *"Cytokine pre-licensed MSCs elicit an enhanced efferocytic response from monocytes leading to improved T-cell suppression,"* **ISCT 2026** Annual Meeting, Dublin, Ireland. (2026, February). Abstract published in ***Cytotherapy***

[5] **D. Chae** and H. Yang, *"Non-viral intron knock-ins unlock splicing-controlled CAR-T cell therapies,"* **K-BioX Summit 2025**, Seoul, Korea. (2025, May). Advisors: Siyeon Rhee and Theodore Roth, Stanford University.

Honors & Awards

President's Award, Korea Institute for Advancement of Technology (KIAT) — **May 2026**
Most outstanding student of the Korea–U.S. High-Tech Industry Youth Exchange Support Program.

Minister's Award, Ministry of Trade, Industry and Energy (MOTIE) — **Jan 2026**
For science communication as a member of the 3rd KORUS 70 Press Corps.

Excellence Award, ResearchThon, AI BioX ConfEX Grand Summit | PHNYX × Cheiron × K-BioX — **Jan 2026**
"Development of an Adapter System to Minimize Air Contamination during Venous Blood Gas (VBG) Sampling."

Final Outstanding Team, B.I.O. Biotechnology Joint Academic Society | Korea University × Yonsei University — **Jun 2025**

Grand Award (Best Poster Presentation), K-BioX Summit 2025 | Yonsei University × Rutgers University × HTAN × G1's Lab — **May 2025**

Outstanding Team, COCF, National Climate Initiative Competition — **Sep 2024**

Skills & Techniques

Cell culture	ucMSC, PBMC and primary monocyte isolation, THP-1 and THP-1-derived macrophages, DC2.4, NR8383, iPSC, cardiomyocytes, adipocyte dedifferentiation
Functional assays	Efferocytosis assays, T-cell suppression (CFSE dilution), macrophage polarization and phenotyping, migration assays, LDH and CCK-8 viability assays
Immunoassays	Flow cytometry and FlowJo, ELISA (Sandwich, Competitive), SDS-PAGE, Western blot, Immunocytochemistry, Bradford and BCA assays, Zeba desalting
Molecular biology	Nucleic acid extraction, RT-PCR and qPCR, primer design, plasmid preparation, electroporation
In vivo	Murine inhalation exposure system operation, necropsy and multi-organ harvest, brain tissue dissection and primary brain cell isolation
Nanomaterials	Core/shell lanthanide scintillating nanoparticles, mesoporous silica and iron oxide nanoparticles, carboxydextran conjugation and surface functionalization, LNP and PEI polyplex formulation, pre-PEGylated NLS-EEP synthesis, DLS/ELS, FT-IR, NanoDrop, ferrozine assay
Imaging	Confocal microscopy, fluorescence live-cell imaging with membrane and organelle trackers, lipid staining
Data Analysis	R (Bioinformatics; bulk RNA-seq, GEO analysis, differential expression, gene module construction), MATLAB, SPSS *ADsP (Advanced Data Analytics Semi-Professional) certified, Korea Data Agency (2025, March).
In silico	Structure-based drug repurposing: AlphaFold-derived receptor structures, hotspot identification, AutoDock Vina virtual screening
Visualization	BioRender, Rhino 3D, ImageJ, Affinity.

Teaching & Extracurricular Activities

Volunteer | World Vision, CHA Gangnam Medical Center, IHCO **Jan 2024 – Present**
Over 120 hours of English–Korean translation for sponsored children's correspondence and workshop leadership for new volunteers; pediatric and emergency department support and monthly rural mobile clinic visits.

Academic Consultant | Modeun Education **Dec 2023 – Present**
Advise high school students on research-based academic portfolios, inquiry report design, personal statements, and admissions interviews; design one-to-one curricula.

Press Corps Member (3rd Cohort, selected nationally) | KORUS 70 / MOTIE **Jul 2025 – May 2026**
Produced video and wrote columns on Korea–U.S. STEM research exchange for a general audience.

Global Ambassador, SIMTOMI | Mediark, medical AI healthcare startup **Aug 2025 – Oct 2025**
Promoted a medical AI application through video content and marketing campaigns built around narrowing gaps in access to care; represented the platform to international student and researcher communities.

Project Coordinator, B.I.O. Joint Academic Society | Korea University and Yonsei University **Feb 2025 – Jun 2025**
Led team BMN2 through an interdisciplinary immunology literature review and discussion project.

Vice President, Environmental Health Science (EHS) Society | Korea University **Mar 2024 – Jan 2025**
Directed the society across the three fields of industrial health and safety, organizing public health campaigns and campus-wide risk assessment projects.

Lead Guitarist and Organizing Committee | TRUSS, Korea University student rock band **Mar 2023 – Jan 2025**
Performed electric guitar in six live shows over two years; organizing committee member for rehearsal scheduling, venue logistics, and stage production.

Mathematics Instructor | Megastudy **May 2022 – Dec 2024**
Taught over 200+ students across 10 schools; developed lesson plans, lecture materials, and visual content for the national e-learning platform.

Climate Policy Researcher, Climate Scouts (9th Cohort) | COCF, non-profit foundation **May 2024 – Sep 2024**
Delivered carbon-reduction education to elementary and middle school students, and enacted a carbon credit policy proposal to the Ministry of Environment.

Scholarships

Udeok Foundation Scholarship | Hanil Cement **Mar 2024 – Graduation**
Merit- and need-based award of KRW 4,500,000 (~USD 3,200) per semester over four semesters (USD 13,000 total)

Korea–U.S. High-Tech Industry Youth Exchange Scholarship (3rd Cohort) | KIAT / MOTIE **May 2025 – May 2026**
Selected on academic merit and potential contribution to Korea's STEM industry (USD 9,000)

Languages

Korean (Native Speaker) | English (TOEFL 5.0/6.0 equivalent to 100/120) | Chinese (HSK Level 2)

References

James Ankrum, Ph.D. · james-ankrum@uiowa.edu
Professor, Roy J. Carver Dept. of Biomedical Engineering, University of Iowa

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